



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

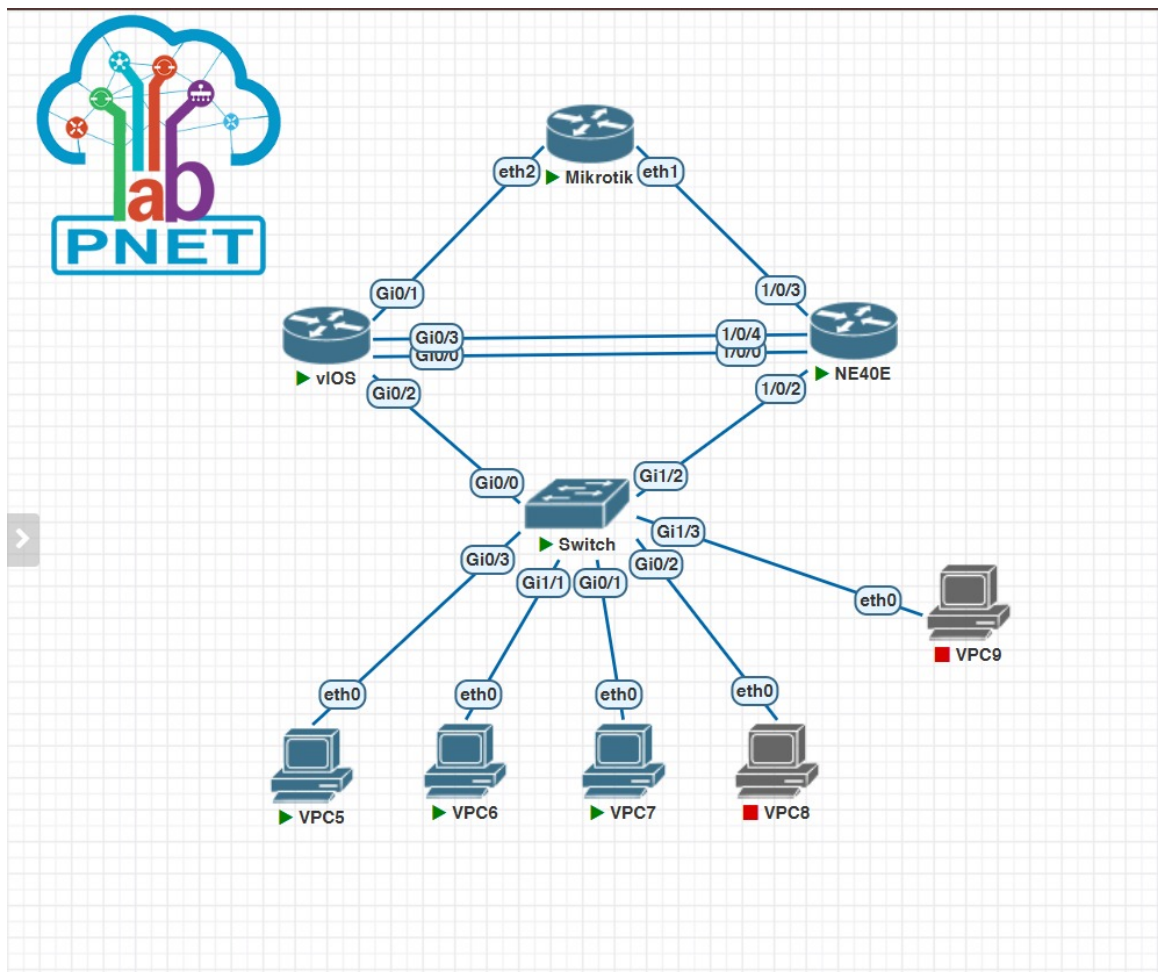
Laporan Akhir Praktikum Jaringan Komputer

Vlan Trunk OSPF MultiVendor

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1 Langkah-Langkah Percobaan



Gambar 1: Topologi Jaringan Praktikum 5

1.1 Praktikum 1: Konfigurasi Cisco Switch (VLAN & Trunking)

1. **Akses Terminal Cisco Switch:** Masuk ke mode *Privileged EXEC* dan masuk ke mode konfigurasi global.
2. **Pembuatan Database VLAN 10 s.d. 40:** Deklarasikan VLAN ID beserta penamaan zonanya (*Finance, HR, IT, Marketingg*) sesuai ketentuan.
3. **Verifikasi Awal Aktivasi VLAN:** Eksekusi perintah `show vlan brief` untuk memastikan seluruh VLAN ID statusnya telah aktif.

```
Terminal
Switch x
Switch#write memory
Building configuration...
Compressed configuration from 3365 bytes to 1648 bytes[OK]
Switch#
*Jun  5 08:41:28.080: %GRUB-5-CONFIG_WRITING: GRUB configuration is being u
pdated on disk. Please wait...
*Jun  5 08:41:28.747: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was writte
n to disk successfully.
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default              active    Gi1/0, Gi1/2, Gi1/3
10   Finance              active    Gi0/2
20   HR                  active    Gi0/1
30   IT                  active    Gi1/1
40   Marketing            active    Gi0/3
1002 fddi-default         act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default     act/unsup
1005 trnet-default       act/unsup
Switch#
```

Gambar 2: Hasil Eksekusi Perintah show vlan brief pada Switch

4. **Alokasi Port Access VLAN 10 (Finance):** Konfigurasi interface gi0/2 menjadi mode access dan masukkan ke dalam VLAN 10.
5. **Alokasi Port Access VLAN 20 (HR):** Atur interface gi0/1 menjadi mode access ke arah VLAN 20 disertai perintah `no shutdown`.
6. **Alokasi Port Access VLAN 30 (IT):** Petakan interface gi1/1 sebagai access port khusus untuk VLAN 30.
7. **Alokasi Port Access VLAN 40 (Marketing):** Daftarkan interface gi0/3 ke dalam keanggotaan access port VLAN 40.
8. **Konfigurasi Enkapsulasi Trunk dot1q:** Terapkan protokol enkapsulasi dot1q pada interface gi0/0 sebelum mengubahnya ke mode trunk.
9. **Aktivasi Mode Trunking ke Core Router:** Set interface gi0/0 menjadi mode trunk dan batasi agar hanya VLAN 10, 20, 30, dan 40 yang diizinkan lewat.
10. **Verifikasi Jalur Trunking Switch:** Jalankan instruksi `show interfaces trunk` untuk memvalidasi pemetaan inter-vlan tagging.

```
Terminal
Switch x
n to disk successfully.
Switch#show interface trunk

Port      Mode           Encapsulation  Status        Native vlan
Gi0/0     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#write memory
Building configuration...
Compressed configuration from 3365 bytes to 1648 bytes[OK]
Switch#
*Jun  5 08:41:28.080: %GRUB-5-CONFIG_WRITING: GRUB configuration is being u
pdated on disk. Please wait...
*Jun  5 08:41:28.747: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was writte
n to disk successfully.
Switch#
```

Gambar 3: Tampilan Output show interfaces trunk Hasil Konfigurasi

11. **Penyimpanan Permanen Konfigurasi:** Tulis konfigurasi berjalan ke memori NVRAM dengan perintah write memory atau copy running-config startup-config.

```
Terminal
Switch x
n to disk successfully.
Switch#show interface trunk

Port      Mode           Encapsulation  Status        Native vlan
Gi0/0     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

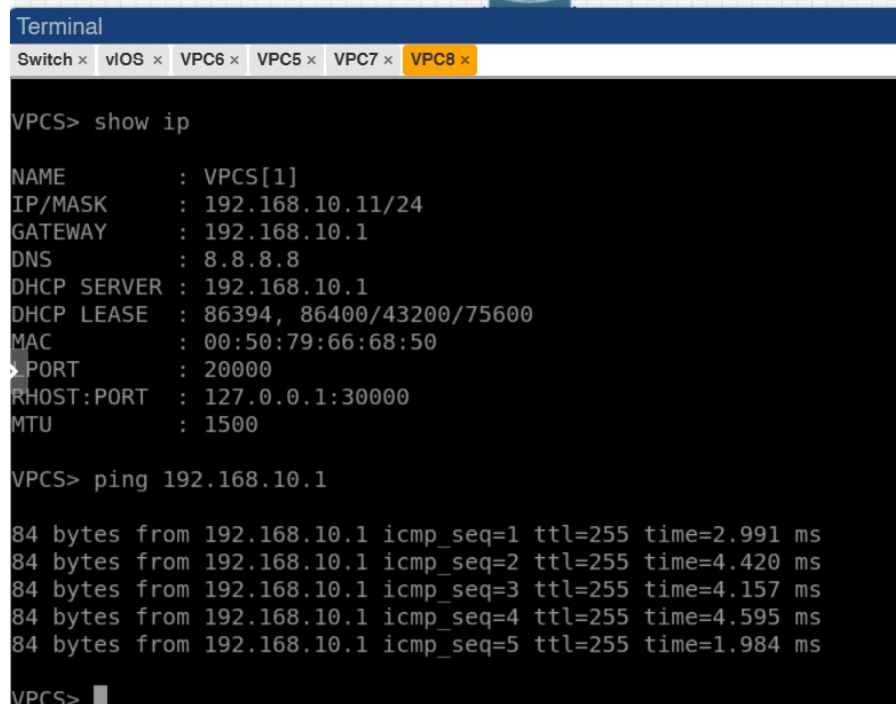
Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#write memory
Building configuration...
Compressed configuration from 3365 bytes to 1648 bytes[OK]
Switch#
*Jun  5 08:41:28.080: %GRUB-5-CONFIG_WRITING: GRUB configuration is being u
pdated on disk. Please wait...
*Jun  5 08:41:28.747: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was writte
n to disk successfully.
Switch#
```

Gambar 4: Proses Commit dan Save Database Konfigurasi Cisco Switch

1.2 Praktikum 2: Konfigurasi Cisco Router (Gateway, ROAS & DHCP)

1. **Aktivasi Physical Interface Trunk:** Buka interface fisik utama gig0/2 yang terhubung ke switch dan hilangkan alamat IP-nya, lalu aktifkan port dengan `no shutdown`.
2. **Konfigurasi Sub-Interface VLAN 10:** Buat sub-interface gig0/2.10, pasang enkapsulasi dot1Q 10, dan definisikan IP Gateway 192.168.10.1.



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86394, 86400/43200/75600
MAC        : 00:50:79:66:68:50
> PORT     : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

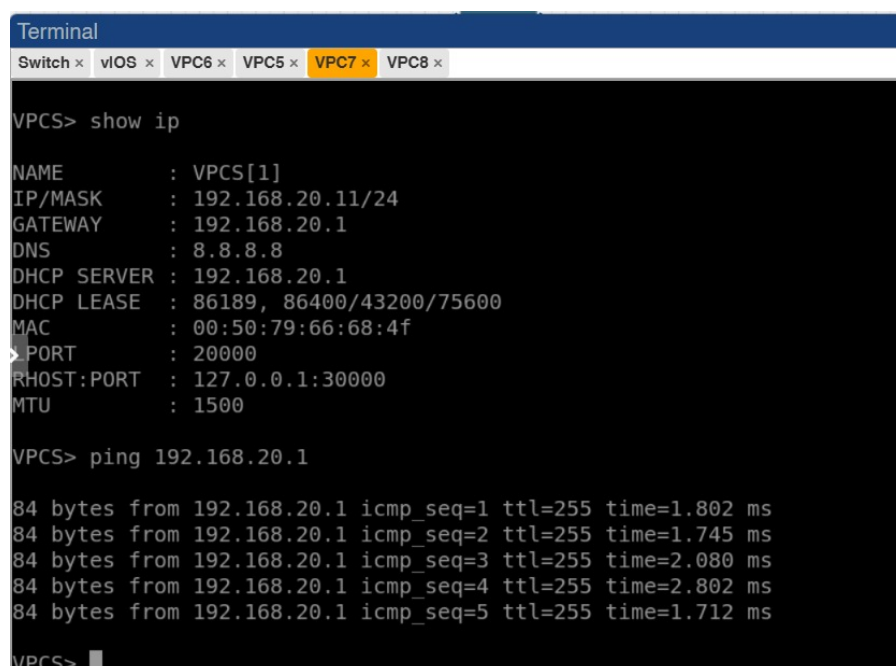
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.991 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=4.420 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=4.157 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=4.595 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.984 ms

VPCS>
```

Gambar 5: Pembuatan Sub-interface dan Gateway untuk Jaringan Finance

3. **Konfigurasi Sub-Interface VLAN 20:** Buat sub-interface gig0/2.20, pasang enkapsulasi dot1Q 20, dan definisikan IP Gateway 192.168.20.1.



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE  : 86189, 86400/43200/75600
MAC        : 00:50:79:66:68:4f
> PORT     : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

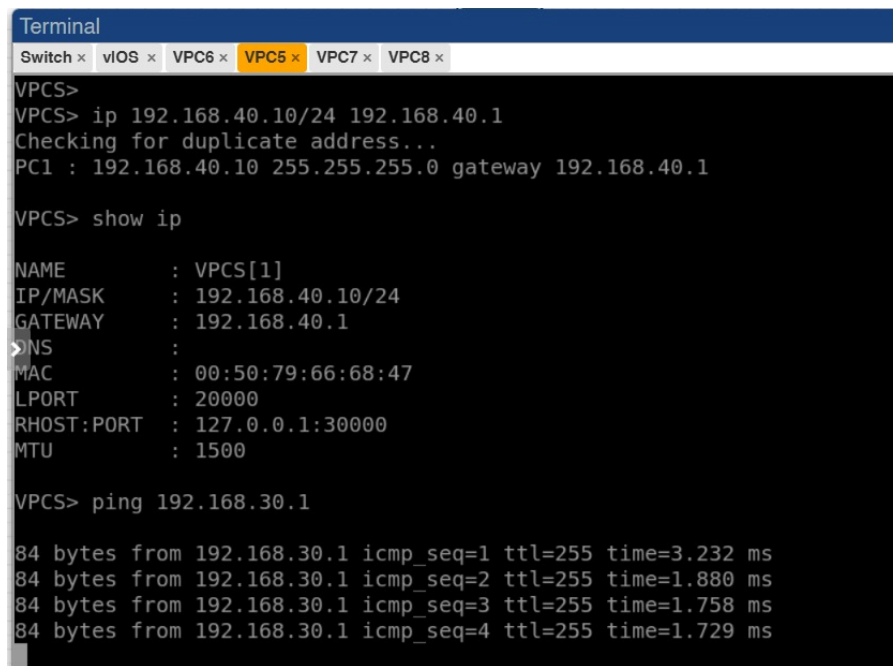
VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.802 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.745 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.080 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.802 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VPCS>
```

Gambar 6: Pembuatan Sub-interface dan Gateway untuk Jaringan HR

4. **Konfigurasi Sub-Interface VLAN 30:** Buat sub-interface gig0/2.30, pasang enkapsulasi dot1Q 30, dan definisikan IP Gateway 192.168.30.1.



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x
VPCS>
VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> show ip

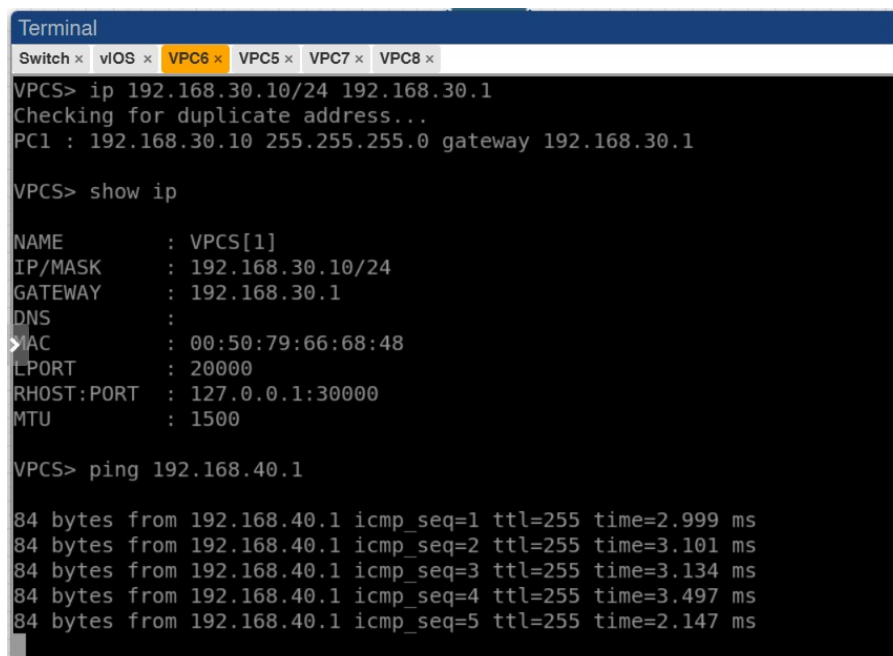
NAME          : VPCS[1]
IP/MASK       : 192.168.40.10/24
GATEWAY       : 192.168.40.1
DNS           :
MAC           : 00:50:79:66:68:47
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.232 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.880 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.758 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.729 ms
```

Gambar 7: Pembuatan Sub-interface dan Gateway untuk Jaringan IT

5. **Konfigurasi Sub-Interface VLAN 40:** Buat sub-interface gig0/2.40, pasang enkapsulasi dot1Q 40, dan definisikan IP Gateway 192.168.40.1.



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip

NAME          : VPCS[1]
IP/MASK       : 192.168.30.10/24
GATEWAY       : 192.168.30.1
DNS           :
MAC           : 00:50:79:66:68:48
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

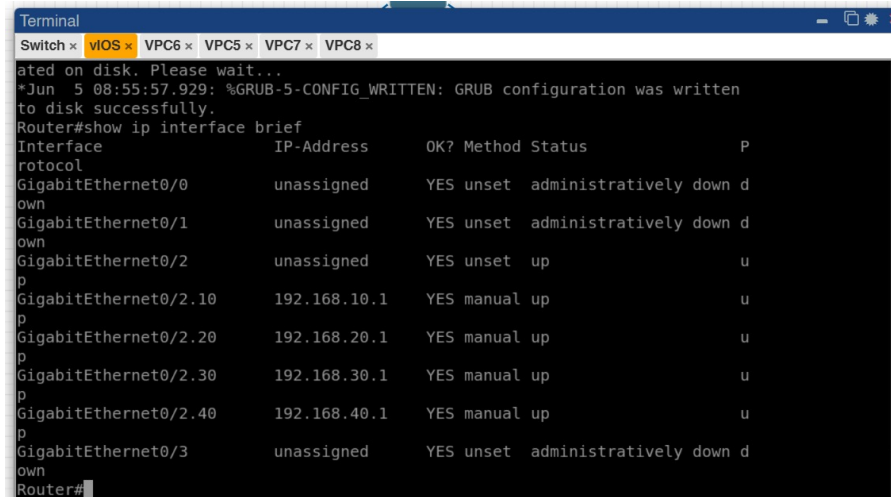
VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=2.999 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.101 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.134 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.497 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.147 ms
```

Gambar 8: Pembuatan Sub-interface dan Gateway untuk Jaringan Marketing

6. **Alokasi Eksklusi IP DHCP Server:** Batasi alokasi alamat IP dinamis dari range .1 hingga .10 agar tidak terjadi bentrok IP (*IP conflict*) pada segmen VLAN 10 dan VLAN 20.
7. **Pembuatan Pool DHCP VLAN 10:** Deklarasikan parameter pool VLAN10-FINANCE lengkap dengan spesifikasi DNS Server 8.8.8.8.

8. **Pembuatan Pool DHCP VLAN 20:** Deklarasikan parameter pool VLAN20-HR lengkap dengan pemetaan gateway utama.
9. **Penyuntikan IP Statis VPCS (VLAN 30 & 40):** Berikan konfigurasi alamat host secara statis pada client IT dan client Marketing via konsol VPCS.
10. **Inspeksi Status Interface Router:** Verifikasi up/down status jalur sub-interface menggunakan perintah `show ip interface brief`.



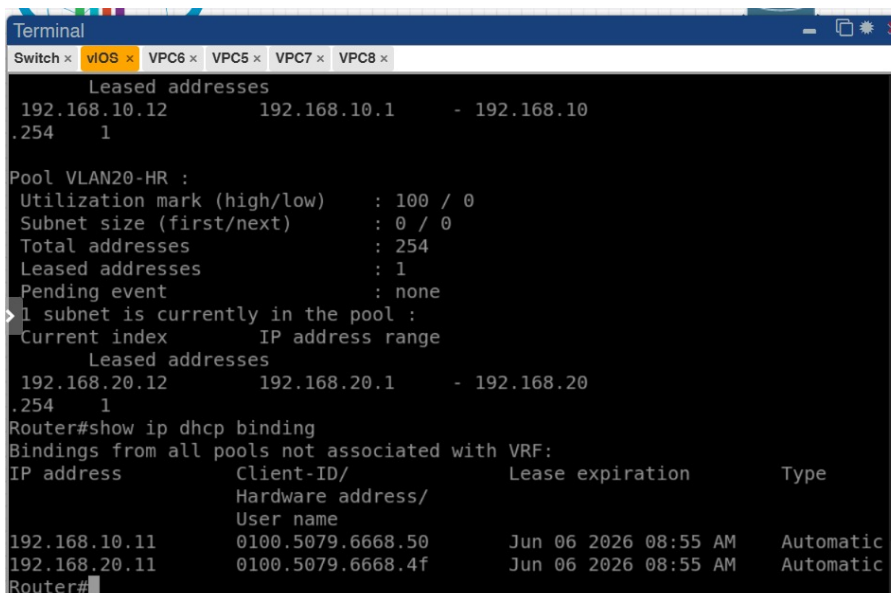
```

Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x
ated on disk. Please wait...
*Jun  5 08:55:57.929: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.
Router#show ip interface brief
Interface                                IP-Address      OK? Method Status      P
GigabitEthernet0/0                      unassigned      YES unset  administratively down d
GigabitEthernet0/1                      unassigned      YES unset  administratively down d
GigabitEthernet0/2                      unassigned      YES unset  up          u
GigabitEthernet0/2.10                   192.168.10.1    YES manual  up          u
GigabitEthernet0/2.20                   192.168.20.1    YES manual  up          u
GigabitEthernet0/2.30                   192.168.30.1    YES manual  up          u
GigabitEthernet0/2.40                   192.168.40.1    YES manual  up          u
GigabitEthernet0/3                      unassigned      YES unset  administratively down d
Router#

```

Gambar 9: Daftar Verifikasi Status Logika Sub-interface Cisco Router

11. **Verifikasi DHCP Lease Pool:** Pastikan data sewa IP dan binding statistik DHCP telah tercatat sempurna di memori router.



```

Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x
Leased addresses
192.168.10.12      192.168.10.1    - 192.168.10
.254      1

Pool VLAN20-HR :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 1
Pending event                     : none
> 1 subnet is currently in the pool :
Current index      IP address range
Leased addresses
192.168.20.12      192.168.20.1    - 192.168.20
.254      1

Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/      Lease expiration      Type
                   Hardware address/
                   User name
192.168.10.11       0100.5079.6668.50   Jun 06 2026 08:55 AM   Automatic
192.168.20.11       0100.5079.6668.4f   Jun 06 2026 08:55 AM   Automatic
Router#

```

Gambar 10: Output Hasil Perintah `show ip dhcp pool` dan `show ip dhcp binding`

12. **Uji Alokasi DHCP IP Dinamis pada Client:** Eksekusi perintah `ip dhcp` pada VPCS VLAN 10 dan VLAN 20 dan tampilkan hasilnya lewat `show ip`.


```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x
Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> ip dhcp
DORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip
>
NAME          : VPCS[1]
IP/MASK        : 192.168.10.11/24
GATEWAY        : 192.168.10.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.10.1
DHCP LEASE     : 86394, 86400/43200/75600
MAC            : 00:50:79:66:68:50
LPORT         : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500

VPCS>
```

Gambar 11: Alokasi IP Otomatis yang Diterima VPCS 10 dan 20

13. **Uji Konektivitas Gateway Lokal:** Lakukan pengujian ping dari seluruh perwakilan PC menuju ke alamat IP sub-interface router-nya masing-masing.

```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x
VPCS> show ip
NAME          : VPCS[1]
IP/MASK        : 192.168.20.11/24
GATEWAY        : 192.168.20.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.20.1
DHCP LEASE     : 86189, 86400/43200/75600
MAC            : 00:50:79:66:68:4f
> PORT         : 20000
RHOST:PORT     : 127.0.0.1:30000
MTU            : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=1.802 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=1.745 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=2.080 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=2.802 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=1.712 ms

VPCS>
```

Gambar 12: Uji Coba Ping Sukses dari Vlan 20 Menuju Default Gateway


```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x
VPCS>
VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.40.10/24
GATEWAY    : 192.168.40.1
DNS        :
MAC        : 00:50:79:66:68:47
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=3.232 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=1.880 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=1.758 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=1.729 ms
```

Gambar 13: Uji Coba Ping Sukses dari Vlan 30 Menuju Default Gateway

```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x
VPCS>
VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.30.10/24
GATEWAY    : 192.168.30.1
DNS        :
MAC        : 00:50:79:66:68:48
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=2.999 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=3.101 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=3.134 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=3.497 ms
84 bytes from 192.168.40.1 icmp_seq=5 ttl=255 time=2.147 ms
```

Gambar 14: Uji Coba Ping Sukses dari Vlan 40 Menuju Default Gateway

14. **Uji Validasi Inter-VLAN Routing:** Lakukan pengetesan komunikasi ping lintas domain (seperti dari PC VLAN 10 ke host VLAN 20/30/40) untuk membuktikan fungsi *Inter-VLAN Routing* berjalan sukses.

```

RHOST:PORT : 127.0.0.1:30000
MTU : 1500

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=2.991 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=4.420 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=4.157 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=4.595 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=1.984 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=3.022 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=4.302 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=2.750 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=3.903 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=2.687 ms
pi
VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=6.091 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=4.045 ms
pin84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=8.408 ms
g 84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.414 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.414 ms

VPCS> ping 192.168.40.10

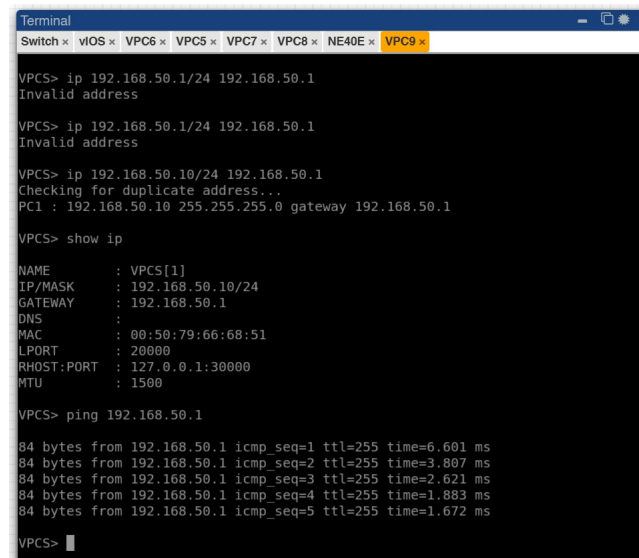
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=2.704 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.930 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=3.793 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=4.196 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=2.828 ms

```

Gambar 15: Hasil Reply Sukses Pengujian Konektivitas Antar Jaringan PC Berbeda VLAN

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

1. **Konfigurasi Segmen LAN Huawei:** Masuk ke system-view, arahkan ke interface Ethernet1/0/2 untuk menaruh gateway 192.168.50.1.



```
Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x

VPCS> ip 192.168.50.1/24 192.168.50.1
Invalid address

VPCS> ip 192.168.50.1/24 192.168.50.1
Invalid address

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:51
LPORT    : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

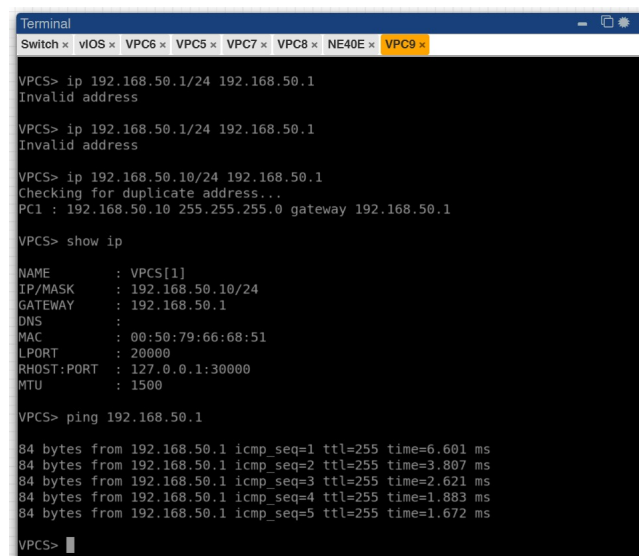
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=6.601 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.807 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.621 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=1.883 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=1.672 ms

VPCS>
```

Gambar 16: Konfigurasi Alamat IP pada Interface Jaringan Lokal Huawei

2. **Pemberian IP Statis PC LAN Huawei:** Konfigurasikan alamat IP manual 192.168.50.10 pada VPCS sisi Huawei dan uji ping ke gateway lokal.



```
Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x

VPCS> ip 192.168.50.1/24 192.168.50.1
Invalid address

VPCS> ip 192.168.50.1/24 192.168.50.1
Invalid address

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS       :
MAC       : 00:50:79:66:68:51
LPORT    : 20000
RHOST:PORT : 127.0.0.1:30000
MTU       : 1500

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=6.601 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.807 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.621 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=1.883 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=1.672 ms

VPCS>
```

Gambar 17: Pengaturan Parameter Jaringan Client Huawei dan Uji Balas Paket Ping

3. **Penyusunan IP Transit Cisco-Huawei Link 1 (10.10.10.0/30):** Berikan alamat IP 10.10.10.1 pada Cisco Gi0/3 dan 10.10.10.2 pada Huawei Ethernet1/0/0. Jangan lupa lakukan commit pada sisi Huawei.
4. **Penyusunan IP Transit Cisco-Huawei Link 2 (10.10.11.0/30):** Berikan parameter alamat IP pada interface Cisco Gi0/0 dan Huawei Ethernet1/0/4.

```
Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
*Jun 5 09:07:31.663: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigabit
Ethernet0/3, changed state to up
Router(config)#inter
Router(config)#interface g0/0
Router(config-if)#desc
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#now shutdown
Router(config-if)#
% Invalid input detected at '^' marker.
Router(config-if)#ex
Router(config)#inter
Router(config)#interface g0/0
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#
*Jun 5 09:09:39.929: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed s
tate to up
*Jun 5 09:09:40.929: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigabit
Ethernet0/0, changed state to up
Router(config)#interface g0/1
Router(config-if)#desc
% Incomplete command.
Router(config-if)#desc
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#
```

Gambar 18: Konfigurasi Alokasi IP Address Transit Link Utama 2 (Cisco - Huawei)

5. **Penyusunan IP Transit Huawei-MikroTik (10.10.20.0/30):** Set IP 10.10.20.1 pada Huawei Ethernet1/0/3 dan tambahkan entri alamat IP 10.10.20.2/30 pada ether1 MikroTik.

```
Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
*Jun 5 09:07:31.663: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigabit
Ethernet0/3, changed state to up
Router(config)#inter
Router(config)#interface g0/0
Router(config-if)#desc
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#now shutdown
Router(config-if)#
% Invalid input detected at '^' marker.
Router(config-if)#ex
Router(config)#inter
Router(config)#interface g0/0
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#
*Jun 5 09:09:39.929: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed s
tate to up
*Jun 5 09:09:40.929: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigabit
Ethernet0/0, changed state to up
Router(config)#interface g0/1
Router(config-if)#desc
% Incomplete command.
Router(config-if)#desc
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#
```

Gambar 19: Konfigurasi Alokasi IP Address Jalur Backup (Huawei - MikroTik)

6. **Penyusunan IP Transit Cisco-MikroTik (10.10.30.0/30):** Pasang IP 10.10.30.1 pada Cisco Gi0/1 dan alamat IP 10.10.30.2/30 pada interface ether2 MikroTik.
7. **Inspeksi Status Interface Global Multi-Vendor:** Jalankan verifikasi show ip interface brief (Cisco), display ip interface brief (Huawei), dan /ip address print (MikroTik).

```

Router(config)#end
Router#write
*Jun  5 09:14:11.891: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Router#
*Jun  5 09:14:14.202: %GRUB-5-CONFIG_WRITING: GRUB configuration is being upd
ated on disk. Please wait...
*Jun  5 09:14:14.902: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.
Router#show ip interface brief
Interface                IP-Address      OK? Method Status    P
GigabitEthernet0/0       10.10.11.1      YES manual up        u
GigabitEthernet0/1       10.10.30.1      YES manual up        u
GigabitEthernet0/2       unassigned      YES unset  up        u
GigabitEthernet0/2.10    192.168.10.1    YES manual up        u
GigabitEthernet0/2.20    192.168.20.1    YES manual up        u
GigabitEthernet0/2.30    192.168.30.1    YES manual up        u
GigabitEthernet0/2.40    192.168.40.1    YES manual up        u
GigabitEthernet0/3       10.10.10.1      YES manual up        u
Router#

```

Gambar 20: Pengecekan Komparatif Tabel IP Address show ip interface brief pada Cisco

```

Router(config)#end
Router#write
*Jun  5 09:14:11.891: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Router#
*Jun  5 09:14:14.202: %GRUB-5-CONFIG_WRITING: GRUB configuration is being upd
ated on disk. Please wait...
*Jun  5 09:14:14.902: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.
Router#show ip interface brief
Interface                IP-Address      OK? Method Status    P
GigabitEthernet0/0       10.10.11.1      YES manual up        u
GigabitEthernet0/1       10.10.30.1      YES manual up        u
GigabitEthernet0/2       unassigned      YES unset  up        u
GigabitEthernet0/2.10    192.168.10.1    YES manual up        u
GigabitEthernet0/2.20    192.168.20.1    YES manual up        u
GigabitEthernet0/2.30    192.168.30.1    YES manual up        u
GigabitEthernet0/2.40    192.168.40.1    YES manual up        u
GigabitEthernet0/3       10.10.10.1      YES manual up        u
Router#

```

Gambar 21: Pengecekan Komparatif Tabel IP Address display ip interface brief pada Huawei

```

Terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
MMM MMM KKK TTTTTTTTT KKK
MMM MMM KKK TTTTTTTTT KKK
MMM MMM III KKK KKK RRRRRR 000000 TTT III KKK KKK
MMM MM III KKKKK RRR RRR 000 000 TTT III KKKKK
MMM MM III KKK KKK RRRRRR 000 000 TTT III KKK KKK
MMM MM III KKK KKK RRR RRR 000000 TTT III KKK KKK

MikroTik RouterOS 6.49.17 (c) 1999-2024 http://www.mikrotik.com/

Do you want to see the software license? [Y/n]: n
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments

[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options

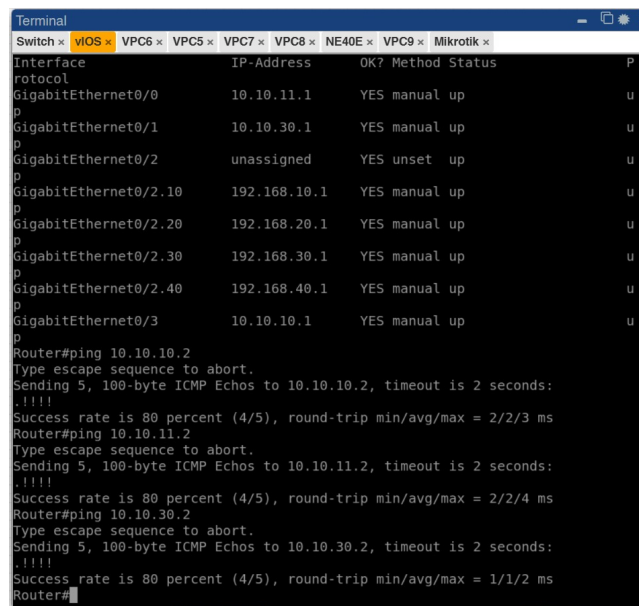
/ Move up to base level
.. Move up one level
/command Use command at the base level

Change your password
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 com
[admin@MikroTik] > 10.10.30.0 ether2
[admin@MikroTik] >
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI 10.10.20.2/30 10.10.20.0 ether1
1 ;; TO-CISCO 10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >

```

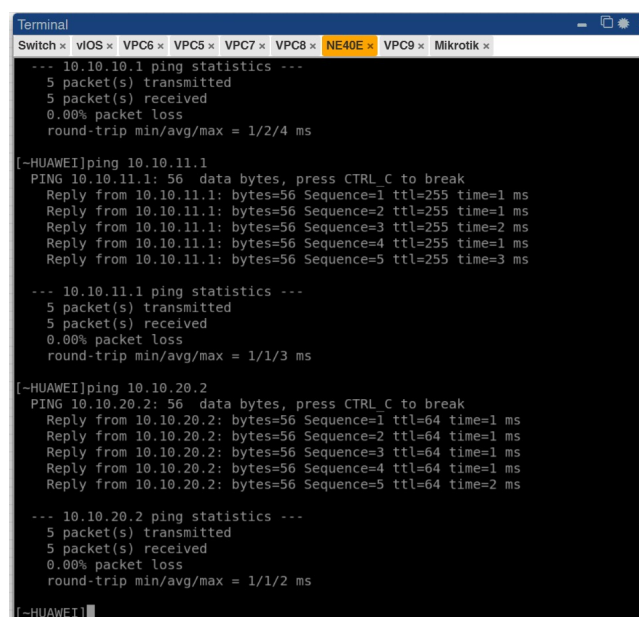
Gambar 22: Pengecekan Komparatif Tabel IP Address /ip address print pada MikroTik

8. **Uji Coba Ping Point-to-Point Antar-Router:** Lakukan cross-ping antar tetangga langsung untuk memastikan interkoneksi Layer 3 aktif sebelum mengaktifkan routing dinamis.



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
Interface IP-Address OK? Method Status
GigabitEthernet0/0 10.10.11.1 YES manual up
GigabitEthernet0/1 10.10.30.1 YES manual up
GigabitEthernet0/2 unassigned YES unset up
GigabitEthernet0/2.10 192.168.10.1 YES manual up
GigabitEthernet0/2.20 192.168.20.1 YES manual up
GigabitEthernet0/2.30 192.168.30.1 YES manual up
GigabitEthernet0/2.40 192.168.40.1 YES manual up
GigabitEthernet0/3 10.10.10.1 YES manual up
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/3 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/4 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#
```

Gambar 23: Hasil Uji Coba Keberhasilan Ping dari Cisco



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
--- 10.10.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/2/4 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=2 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/3 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]
```

Gambar 24: Hasil Uji Coba Keberhasilan Ping dari Huawei

```
10.10.30.2/30      10.10.30.0      ether2
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                SIZE TTL TIME  STATUS
0 10.10.20.1            56 255 3ms
1 10.10.20.1            56 255 1ms
2 10.10.20.1            56 255 0ms
3 10.10.20.1            56 255 1ms
4 10.10.20.1            56 255 0ms
5 10.10.20.1            56 255 0ms
6 10.10.20.1            56 255 1ms
7 10.10.20.1            56 255 0ms
8 10.10.20.1            56 255 0ms
sent=9 received=9 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=3ms

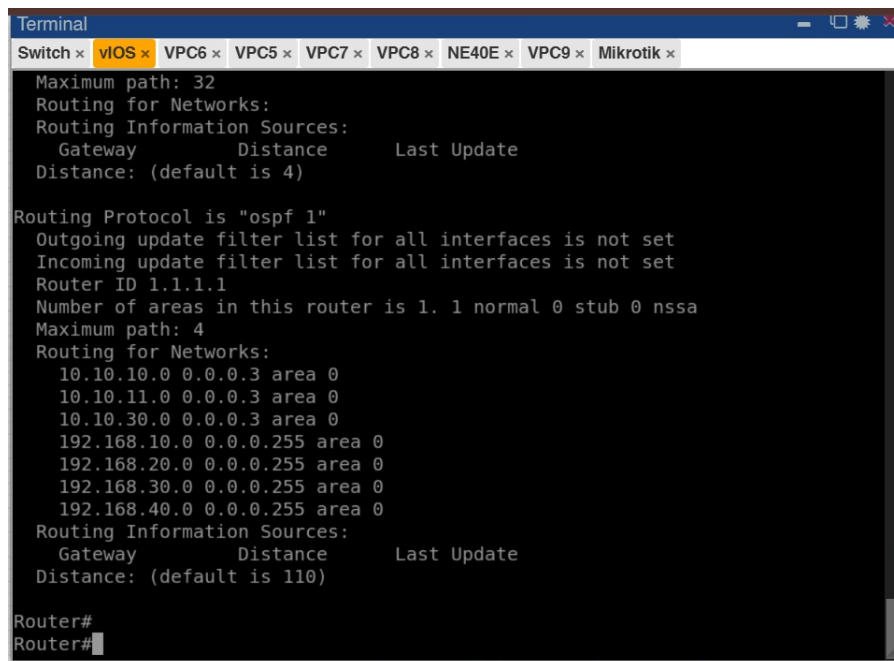
[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                SIZE TTL TIME  STATUS
0 10.10.30.1            56 255 1ms
1 10.10.30.1            56 255 1ms
2 10.10.30.1            56 255 1ms
3 10.10.30.1            56 255 1ms
4 10.10.30.1            56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=1ms

[admin@MikroTik] >
```

Gambar 25: Hasil Uji Coba Keberhasilan Ping dari Mikrotik

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

1. **Konfigurasi Routing OSPF Sisi Cisco Router:** Aktifkan modul `router ospf 1`, tetapkan router-id 1.1.1.1, lalu daftarkan seluruh segmen network transit inter-router beserta keempat subnet internal VLAN (VLAN 10, 20, 30, dan 40) ke dalam Area 0 menggunakan wildcard mask.



```
Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x

Maximum path: 32
Routing for Networks:
Routing Information Sources:
  Gateway      Distance      Last Update
Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway      Distance      Last Update
  Distance: (default is 110)

Router#
Router#
```

Gambar 26: Inisialisasi Proses OSPF dan Pengiklanan Subnet Network pada Cisco Router

2. **Manipulasi Metric Cost OSPF Cisco:** Masuk ke parameter konfigurasi interface transit utama (gi0/3 dan gi0/0) untuk menetapkan nilai `ospf cost 10`. Sebaliknya, set nilai `ospf cost 100` pada interface jalur cadangan (gi0/1) menuju MikroTik.
3. **Konfigurasi Routing OSPF Sisi Huawei Router:** Masuk ke tampilan sistem (*system-view*) Huawei, jalankan perintah `ospf 1 router-id 2.2.2.2`, aktifkan area 0.0.0.0, kemudian daftarkan seluruh segmen network point-to-point terhubung beserta subnet lokal LAN Huawei (192.168.50.0).

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.10.1       BDR: 10.10.10.2     MTU: 1500
  Dead timer due in 38 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m08s
  Neighbor Up Time : 2026-06-05 09:31:07
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full           Mode:Nbr is Slave   Priority: 1
  DR: 10.10.11.1       BDR: 10.10.11.2     MTU: 1500
  Dead timer due in 37 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m08s
  Neighbor Up Time : 2026-06-05 09:31:07
  Authentication Sequence: [ 0 ]

[~HUAWEI]

```

Gambar 27: display ospf peer

```

Terminal
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x

0/4
0/0
  192.168.30.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/
0/4
0/0
  192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/
0/4
0/0
  192.168.50.0/24 OSPF 10 11 D 10.10.10.1 Ethernet1/
0/0

OSPF routing table status : <Inactive>
      Destinations : 4      Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
0/0 10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/
0/4 10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/
0/3 10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/
0/2 192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/

[~HUAWEI]

```

Gambar 28: display ip routing-table protocol ospf

```

terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0
.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0
.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0
.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0
.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0
.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0
.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]

```

Gambar 29: display ospf routing

4. **Manipulasi Metric Cost OSPF Huawei:** Modifikasi parameter biaya jalur pada interface Ethernet1/0/0 dan Ethernet1/0/4 dengan nilai cost 10, serta berikan nilai cost 100 pada port backup Ethernet1/0/3.
5. **Konfigurasi Routing OSPF Sisi MikroTik Router:** Akses terminal MikroTik, tetapkan parameter router-id instans default menjadi 3.3.3.3. Selanjutnya, masukkan network link transit (10.10.20.0/30 dan 10.10.30.0/30) ke dalam database area tulang punggung (*backbone*).

```

terminal
Switch x VIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
SEQ HOST SIZE TTL TIME STATUS
0 10.10.30.1 56 255 1ms
1 10.10.30.1 56 255 1ms
2 10.10.30.1 56 255 1ms
3 10.10.30.1 56 255 1ms
4 10.10.30.1 56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=1ms

[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2
priority=1 dr-address=10.10.30.1 backup-dr-address=10.10.30.2
state="Full" state-changes=5 ls-retransmits=0 ls-requests=0
db-summaries=0 adjacency=12s
1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1
priority=1 dr-address=10.10.20.1 backup-dr-address=10.10.20.2
state="Full" state-changes=5 ls-retransmits=0 ls-requests=0
db-summaries=0 adjacency=19s
[admin@MikroTik] >

```

Gambar 30: Konfigurasi Router-ID dan Network OSPF Backbone Menggunakan CLI MikroTik

6. **Manipulasi Metric Cost OSPF MikroTik:** Naikkan nilai biaya hitung (*cost*) pada interface ether1 dan ether2 MikroTik secara manual menjadi 100 agar posisinya stabil bertindak sebagai link redundansi cadangan.

```

[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2
  priority=1 dr-address=10.10.30.1 backup-dr-address=10.10.30.2
  state="Full" state-changes=5 ls-retransmits=0 ls-requests=0
  db-summaries=0 adjacency=12s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1
  priority=1 dr-address=10.10.20.1 backup-dr-address=10.10.20.2
  state="Full" state-changes=5 ls-retransmits=0 ls-requests=0
  db-summaries=0 adjacency=19s
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#    DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.30.1 110
      10.10.20.1
1 ADo 10.10.11.0/30      10.10.30.1 110
      10.10.20.1
2 ADo 192.168.10.0/24    10.10.30.1 110
3 ADo 192.168.20.0/24    10.10.30.1 110
4 ADo 192.168.30.0/24    10.10.30.1 110
5 ADo 192.168.40.0/24    10.10.30.1 110
6 ADo 192.168.50.0/24    10.10.20.1 110
[admin@MikroTik] >

```

Gambar 31: Pemberian Nilai Metric Cost Backup pada Port Interface MikroTik

7. **Verifikasi Pembentukan OSPF Neighbor Multi-Vendor:** Lakukan inspeksi relasi kedekatan adjacency tetangga pada masing-masing router. Pastikan Cisco Router dan Huawei Router sukses membentuk status *FULL* neighbor dengan MikroTik beserta tautan inter-koneksi utamanya.

```

B - blackhole, U - unreachable, P - prohibit
#    DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 ADo 10.10.10.0/30      10.10.30.1 110
      10.10.20.1
1 ADo 10.10.11.0/30      10.10.30.1 110
      10.10.20.1
2 ADo 192.168.10.0/24    10.10.30.1 110
3 ADo 192.168.20.0/24    10.10.30.1 110
4 ADo 192.168.30.0/24    10.10.30.1 110
5 ADo 192.168.40.0/24    10.10.30.1 110
6 ADo 192.168.50.0/24    10.10.20.1 110
[admin@MikroTik] > show ip ospf neighbor
bad command name show (line 1 column 1)
[admin@MikroTik] > show ip ospf neighbor
bad command name show (line 1 column 1)
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2
  priority=1 dr-address=10.10.30.1 backup-dr-address=10.10.30.2
  state="Full" state-changes=5 ls-retransmits=0 ls-requests=0
  db-summaries=0 adjacency=2m57s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1
  priority=1 dr-address=10.10.20.1 backup-dr-address=10.10.20.2
  state="Full" state-changes=5 ls-retransmits=0 ls-requests=0
  db-summaries=0 adjacency=3m4s
[admin@MikroTik] >

```

Gambar 32: Tampilan Output Pengecekan Status OSPF Adjacency Neighbor / Peer

8. **Inspeksi Tabel Routing Dinamis OSPF Global:** Eksekusi perintah pembacaan tabel rute pada ketiga perangkat. Pastikan Cisco berhasil mendapatkan route OSPF menuju LAN Huawei (192.168.50.0/24), Huawei mendapatkan route menuju keempat segmen VLAN Cisco, serta MikroTik mendapatkan semua entri informasi rute network.

```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
public Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.30.0/30       OSPF   10   110      D    10.10.11.1    Ethernet1/0/4
                  OSPF   10   110      D    10.10.10.1    Ethernet1/0/0
192.168.10.0/24      OSPF   10   11        D    10.10.11.1    Ethernet1/0/4
                  OSPF   10   11        D    10.10.10.1    Ethernet1/0/0
192.168.20.0/24      OSPF   10   11        D    10.10.11.1    Ethernet1/0/4
                  OSPF   10   11        D    10.10.10.1    Ethernet1/0/0
192.168.30.0/24      OSPF   10   11        D    10.10.11.1    Ethernet1/0/4
                  OSPF   10   11        D    10.10.10.1    Ethernet1/0/0
192.168.40.0/24      OSPF   10   11        D    10.10.11.1    Ethernet1/0/4
                  OSPF   10   11        D    10.10.10.1    Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost    Flags NextHop        Interface
-----
10.10.10.0/30       OSPF   10   10        10.10.10.2    Ethernet1/0/0
10.10.11.0/30       OSPF   10   10        10.10.11.2    Ethernet1/0/4
10.10.20.0/30       OSPF   10  100        10.10.20.1    Ethernet1/0/3
192.168.50.0/24     OSPF   10   1         192.168.50.1    Ethernet1/0/2
[~HUAWEI]
```

Gambar 33: Hasil Verifikasi Pertukaran Tabel Rute Antar-Router Multi-Vendor

- Uji Validasi Konektivitas End-to-End PC Client:** Jalankan pengujian komunikasi ping dua arah secara berkala. Pastikan seluruh host PC dari VLAN 10, 20, 30, dan 40 sukses melakukan *PING* ke PC LAN Huawei, dan sebaliknya PC LAN Huawei dapat memberikan respons balas balik (*Reply*) dengan sempurna.

```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=9.253 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.632 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=5.682 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.414 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.177 ms

VPCS>
```

Gambar 34: Hasil Reply Sukses Pengujian Konektivitas Ping PC VLAN 10 dapat ping PC LAN Huawei.

```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
V/
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.632 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.169 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.238 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.120 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.570 ms

VPCS>
```

Gambar 35: Hasil Reply Sukses Pengujian Konektivitas Ping PC VLAN 20 dapat ping PC LAN Huawei.

```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
VPCS>
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=8.052 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.859 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.066 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=7.381 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=6.129 ms
VPCS>
```

Gambar 36: Hasil Reply Sukses Pengujian Konektivitas Ping PC VLAN 30 dapat ping PC LAN Huawei.

```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
VPCS> ping 192.168.50.10
84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=2.956 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.890 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.343 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.413 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.579 ms
VPCS>
VPCS>
```

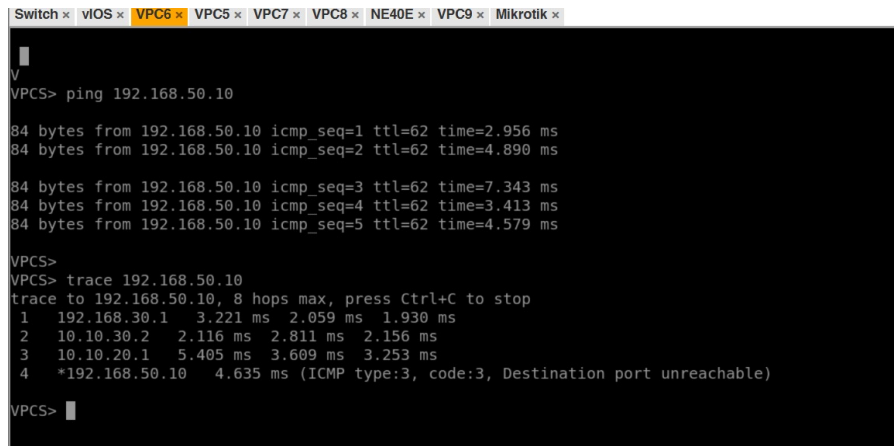
Gambar 37: Hasil Reply Sukses Pengujian Konektivitas Ping PC VLAN 40 dapat ping PC LAN Huawei.

```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
VPCS> ping 192.168.10.11
84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=10.006 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=2.891 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=7.118 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=5.702 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=3.918 ms
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=4.165 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=4.539 ms
VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=4.183 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=3.926 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.087 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=7.342 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=4.647 ms
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=4.334 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=3.179 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=4.446 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=3.352 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=5.355 ms
VPCS>
```

Gambar 38: Hasil Reply Sukses Pengujian Konektivitas Ping Lintas Jaringan PC Client

1.5 Praktikum 5: Pengujian Failover dan Redundansi OSPF

1. **Traceroute Saat Dua Link Utama Aktif (Kondisi Normal):** Lakukan pengujian *traceroute* dari PC Client menuju LAN Huawei (192.168.50.10) saat kondisi seluruh link inter-koneksi menyala. Amati bahwa paket melewati jalur langsung (*direct Link 1 / Link 2*) antara Cisco dan Huawei.



```
Switch x vIOS x VPC6 x VPC5 x VPC7 x VPC8 x NE40E x VPC9 x Mikrotik x
V
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=2.956 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.890 ms

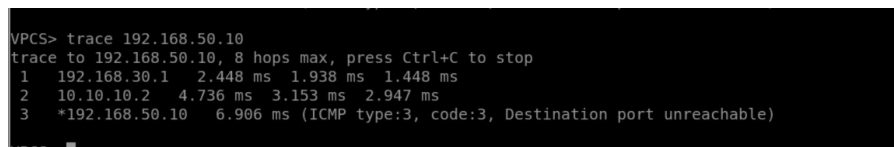
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.343 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.413 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.579 ms

VPCS>
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.30.1    3.221 ms  2.059 ms  1.930 ms
 2  10.10.30.2     2.116 ms  2.811 ms  2.156 ms
 3  10.10.20.1     5.405 ms  3.609 ms  3.253 ms
 4  *192.168.50.10 4.635 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 39: Screenshot Hasil Trace 192.168.50.10 Saat Dua Link Utama Aktif

2. **Traceroute Saat Satu Link Utama Mati (Link Failure Fase 1):** Matikan salah satu link utama Cisco-Huawei (misal *gi0/3*) melalui perintah *shutdown*. Jalankan kembali perintah *trace* dari PC Client; paket data harus terpantau tetap melewati jalur Cisco-Huawei menggunakan sisa link utama yang masih aktif (10.10.11.2).

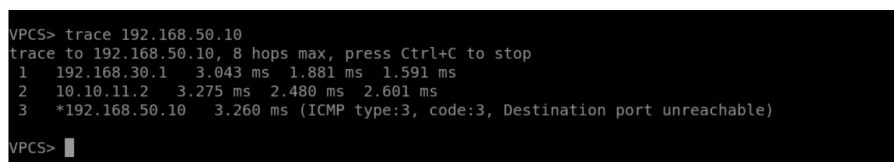


```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.30.1    2.448 ms  1.938 ms  1.448 ms
 2  10.10.10.2     4.736 ms  3.153 ms  2.947 ms
 3  *192.168.50.10 6.906 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 40: Screenshot Hasil Trace 192.168.50.10 Saat Satu Link Mati

3. **Traceroute Melewati MikroTik (Kondisi Dua Link Utama Mati):** Lakukan *shutdown* pada link utama kedua (*gi0/0*) sehingga jalur utama putus total. Eksekusi perintah *trace* dari client dan pastikan algoritma OSPF telah berhasil melakukan *failover* dengan mengalihkan jalur melintasi simpul rute cadangan MikroTik.



```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.30.1    3.043 ms  1.881 ms  1.591 ms
 2  10.10.11.2     3.275 ms  2.480 ms  2.601 ms
 3  *192.168.50.10 3.260 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
```

Gambar 41: Screenshot Hasil Trace 192.168.50.10 yang Menunjukkan Jalur Melewati MikroTik

2 Hasil Tugas Modul

Tugas modul ini berfokus pada implementasi arsitektur jaringan *Enterprise HQ-Branch* yang menghubungkan kantor pusat (Jakarta) dan kantor cabang (Surabaya). Seluruh alur komunikasi antar-site disatukan menggunakan teknologi **tunneling GRE (Generic Routing Encapsulation)** yang berjalan

di atas topologi fisik berbasis FortiGate. Pada tingkat manajemen rute, protokol **OSPF** diaktifkan di atas *tunnel* untuk mendistribusikan informasi jaringan secara dinamis, dilengkapi dengan mekanisme *redistribute static route* guna memastikan subnet internal di kedua sisi dapat saling dikenal.

Sisi Jakarta menerapkan mekanisme *high availability* menggunakan **VRRP (Virtual Router Redundancy Protocol)**, yang melibatkan sinkronisasi *gateway* antara Cisco Router dan MikroTik Router. Layanan alamat IP pusat dikoordinasikan secara terpusat oleh **Ubuntu Server** melalui protokol **ISC-DHCP**, di mana permintaan klien dari VLAN yang berbeda diteruskan melalui *DHCP Relay* pada perangkat *gateway*.

Sisi cabang di Surabaya menggunakan pendekatan *firewall-centric* dengan FortiGate bertindak sebagai *edge* keamanan, yang diintegrasikan dengan MikroTik sebagai manajemen VLAN lokal. Berdasarkan serangkaian pengujian konektivitas *end-to-end*, sistem telah berhasil mencapai integrasi fungsional penuh, mulai dari alokasi IP dinamis, akses internet terpusat melalui NAT *gateway*, hingga aksesibilitas layanan *web server* Jakarta dari *client* cabang Surabaya melalui jalur *tunnel* terproteksi.

3 Dokumentasi Tugas Modul

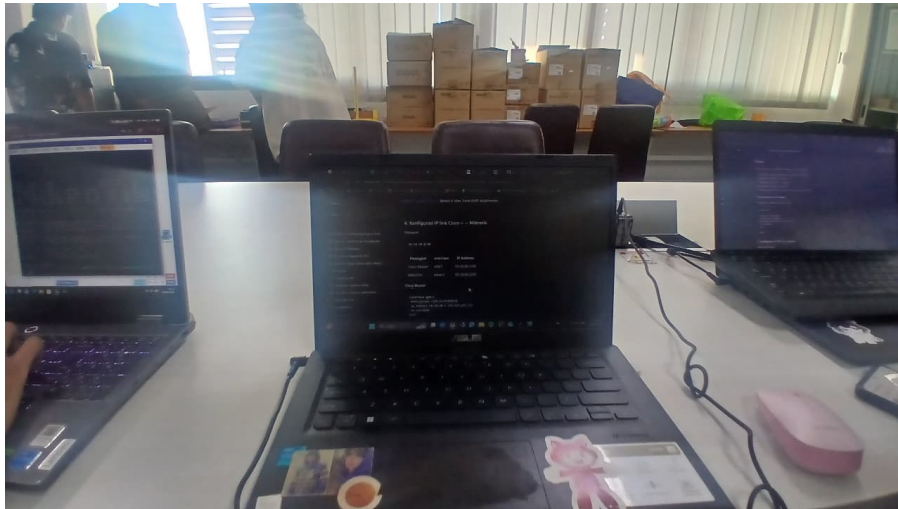
Dokumentasi visual mengenai tahapan konfigurasi dan hasil pengujian *ping* (termasuk *traceroute* serta *routing table*) untuk setiap sub-modul telah didokumentasikan. Seluruh *screenshot* pendukung dapat dilihat pada file `README.md` di repositori GitHub praktikum terkait.

4 Kesimpulan

Praktikum Modul 5 ini berhasil memberikan pemahaman mendalam mengenai integrasi teknologi switching Layer 2 dan sistem routing dinamis Layer 3 pada lingkungan komputasi heterogen berbasis perangkat **multi-vendor**. Praktikan mampu menganalisis pentingnya manajemen VLAN dan alokasi trunking dalam mereduksi beban domain *broadcast* lokal. Di samping itu, simulasi pengubahan metrik **OSPF Cost** dan pemutusan link fisik memberikan bukti empiris mengenai keandalan algoritma Shortest Path First (SPF) Dijkstra dalam membentuk ekosistem jaringan yang responsif, adaptif, serta memiliki fitur redundansi otomatis (**failover**) yang andal terhadap risiko kegagalan link (**link failure**).

5 Lampiran

5.1 Dokumentasi saat praktikum



Gambar 42: Dokumentasi Kegiatan Praktikum VLAN-Trunk-OSPF Kelompok 21